Development of Computer Aided Diagnostic System for Preliminary Medical Image Screening

A PROJECT REPORT

Submitted

in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Submitted by

1. Ankita Kumari (22104132014)
2. Kirti Krishna (22104132015)
3. Tanu Shri (22104132022)
4. Ankit Kumar (23104132904)
5. Priti Kumari (23104132905)





Department of Electronics and Communication Engineering

Saharsa College of Engineering, Saharsa

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Under the Supervision of

Md. Tabish Raza
Assistant Professor





Department of Electronics and Communication Engineering
Saharsa College of Engineering, Saharsa
Bihar Engineering University, Patna

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Sl. No.	Name of Student	Registration No.	Signature
01.	Ankita Kumari	22104132014	
02.	Kirti Krishna	22104132015	
03.	Tanu Shri	22104132022	
04.	Ankit Kumar	23104132904	
05.	Priti Kumari	23104132905	

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This is to certify that project report entitled "**DEVELOPMENT OF COMPUTER AIDED DIAGNOSTIC SYSTEM FOR PRELIMINARY MEDICAL IMAGE SCREENING**" which is submitted by "Ankita Kumari, Kirti Krishna, Tanu Shri, Ankit Kumar, Priti Kumari" in partial fulfilment of the requirement, for the 06 credit course Project-II (Course code-100801), of the degree of Bachelor of Technology in department of Electronics & Communication Engineering of Saharsa College of Engineering, Saharsa is a record of the candidate own work carried out by them under my/our supervision. The work matter embodied in this project report is original and has not been submitted anywhere for the award of any other degree or diploma.

(Prof. Md. Tabish Raza) Project Supervisor SCE Saharsa	(Head of Department) Dept. of Electronics & Comm Engg. SCE Saharsa	(External Examiner)

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Ankita Kumari - 22104132014

Kirti Krishna - 22104132015

Tanu Shri - 22104132022

Ankit Kumai - 23104132904

Priti Kumari - 23104132905

ABSTRACT

Medical image segmentation plays a vital role in modern healthcare systems by assisting in disease diagnosis, treatment planning, and medical image analysis. The increasing availability of medical imaging modalities such as MRI, CT scan, ultrasound, and X-ray has created the need for automated and accurate image processing techniques.

This project presents the development of a Computer Aided Diagnostic (CAD) system for preliminary medical image screening using image segmentation techniques. The proposed work focuses on preprocessing, thresholding, clustering, and segmentation of medical images to identify meaningful regions such as tissues, tumors, and abnormal structures.

Various segmentation methods including Thresholding, Edge-based Segmentation, Region-based Segmentation, Watershed Segmentation, and K-Means Clustering have been studied and implemented. Among these methods, Modified OTSU Thresholding and K-Means Clustering provided better segmentation performance for different medical image datasets.

The system was implemented using MATLAB/Python-based image processing tools. Performance evaluation was carried out using standard image quality metrics such as PSNR, Dice Coefficient, Jaccard Index, Accuracy, and F1-Score.

Experimental results demonstrate that the proposed segmentation approach improves image quality and provides reliable segmentation performance for preliminary medical screening applications. The developed system can assist healthcare professionals in faster and more accurate medical diagnosis.

Keywords: Medical Image Segmentation, OTSU Thresholding, K-Means Clustering, CAD System, MRI, Medical Imaging, Digital Image Processing.

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CHAPTER 1

1.INTRODUCTION

1.1 BASIC OF DIGITAL IMAGE PROCESSING

An image may be defined as a two-dimensional function, f (x, y), where x and y are spatial (plane) coordinates, and the amplitude of at any pair of coordinates (x, y) is called the intensity or gray level of the image at that point. When x, y, and the intensity values of f are all finite, discrete quantities, we call the image a digital image. The field of digital image processing refers to processing digital images by means of a digital computer. Note that a digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are called picture elements, image elements, pels, and pixels. Pixel is the term used most widely to denote the elements of a digital image.[1]

Digital image processing may be classified into various subbranches based on method whose:

i. Input and output are image and

ii. Inputs may be images whereas outputs are attributes extracted from those images.

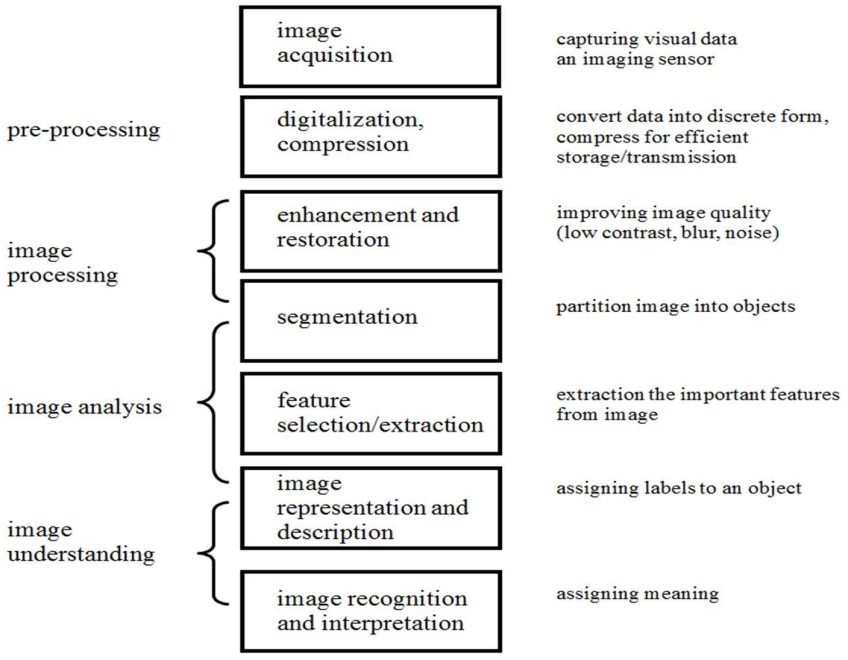


Figure 1. Digital image processing scheme

1.2 DIGITAL IMAGE AND ITS TYPE

Image

An image in "real world" is considered to be a function of two real variables, for example a $f(x, y)$ with a as the amplitude of the image at a real co-ordination position (x, y). An image is a two-dimensional function $f(x, y)$, $f(x, y)$ where x and y are the spatial(plane) coordinates and the amplitude off at any pair of coordinates (x, y). is called the intensity of the image at the level.[1]

Digital Image

A digital image is an image a (x, y). that has been discretized both in spital co-ordinates and brightness. Digitization of spatial co-ordinates (x, y). is called image sampling. Amplitude digitization is called gray level quantization. Modern digital technology manipulates multi-dimensional signals with systems that range from simple digital circuit to advanced parallel computer.[1]

Types of Images

There are several standard types of images. They are

1. Binary
2. Gray Scale
3. Color

1. Binary Image

Binary images are the simplest type of images and can take two values, typically black and white, 0 and 1. A binary image is referred to as a 1-bit image because it takes only 1 binary digit to represent each pixel. These types of images are frequently used in application where the only information required is general shape or outline.[1]

- **0 → Black**
- **1 → White**

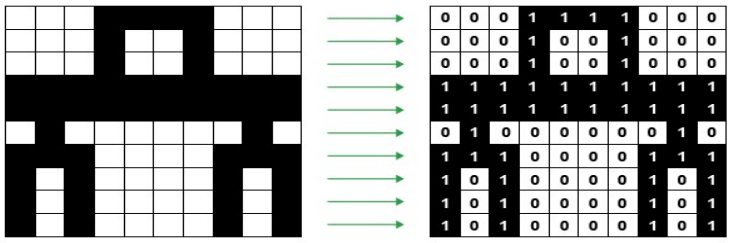


Figure 2. Example of binary image

2. Grayscale Image

Gray scale image is referred to monochrome (one-color) images. They contain gray-level information, no color information. The number of bits used for each pixel determine the number of different gray-levels available. They typically gray-scale contains 8-bit/pixel

data, which allow us to have 256 different gray level. It contains 256 intensity levels (0 - 255).[1]

0 → Black, 1→ White

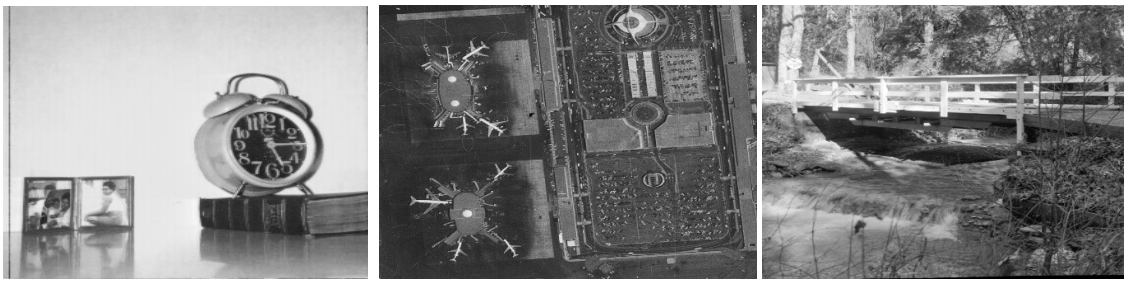


Figure 3. Example of gray scale image

3. Color image

Color image can be modeled as three-band monochrome image date, where each band of data corresponds to a different color. The actual information stored in the digital image data is the gray level information in spectral band.[1]

Typicaly color image are represented by RGB image.

R → Red

G → Green

B → Blue

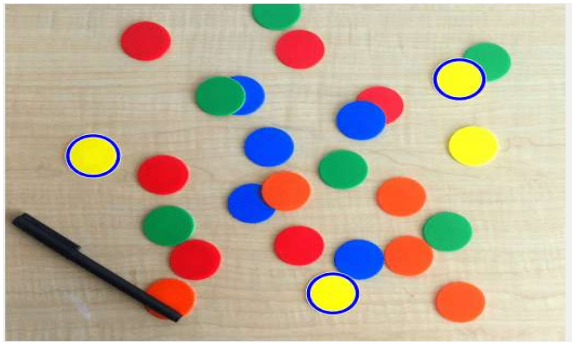


Figure 4. Example of Color image

1.3 TOOLS OF DIGITAL IMAGE PROCESSING

Digital image processing is an important field that deals with the manipulation and analysis of digital image using computer-based techniques. To perform different image processing operation like enhancement, filtering, segmentation, and recognition, various software and hardware tools are used.

Some commonly used tools are

i. MATLAB

MATLAB is widely preferred in academics because it provides a dedicated *Image Processing Toolbox*. With MATLAB, it becomes very easy to perform operations such as edge detection, image filtering, and transformation. It also allows visualization of results, which helps in better understanding.[2]

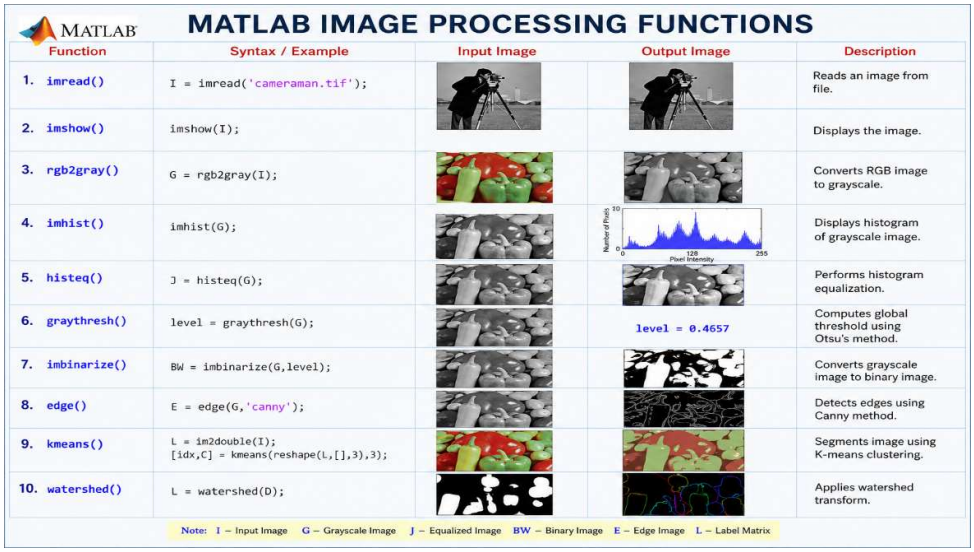


Figure 5. MATLAB function

ii. Python

Python is open-source and very popular nowadays. Python provides powerful libraries like OpenCV, NumPy, and Matplotlib. OpenCV is mainly used for real-time image processing, while NumPy helps in mathematical operations and Matplotlib is used for displaying images. Due to its simplicity and flexibility, Python is widely used in both research and industry.[3]

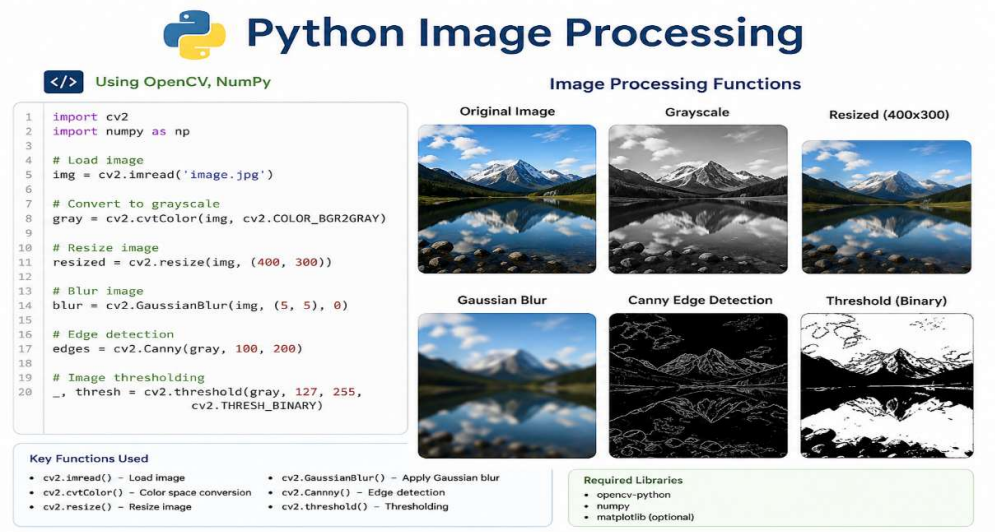


Figure 6. Python

iii. Open CV (Open-Source Computer Vision Library)

Open CV is itself a very powerful tool used for computer vision tasks. It supports multiple programming languages and is mainly used in applications like face detection, object tracking, and video processing. It is fast and efficient, especially for real-time systems.[3]

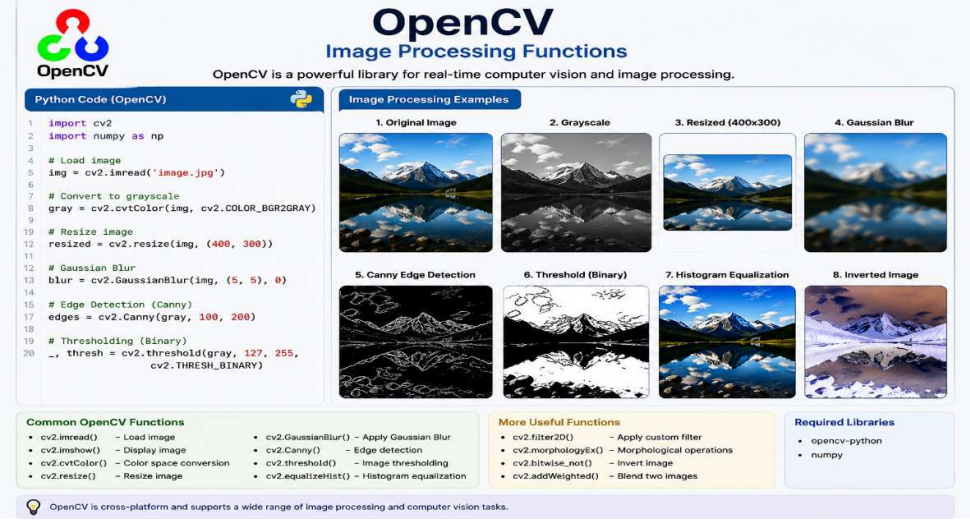


Figure 7. Open CV

iv. ImageJ

ImageJ is an open-source software mostly used in scientific and medical image analysis. It supports plugins and can handle multi-dimensional images, making it very useful in biomedical applications.[4]

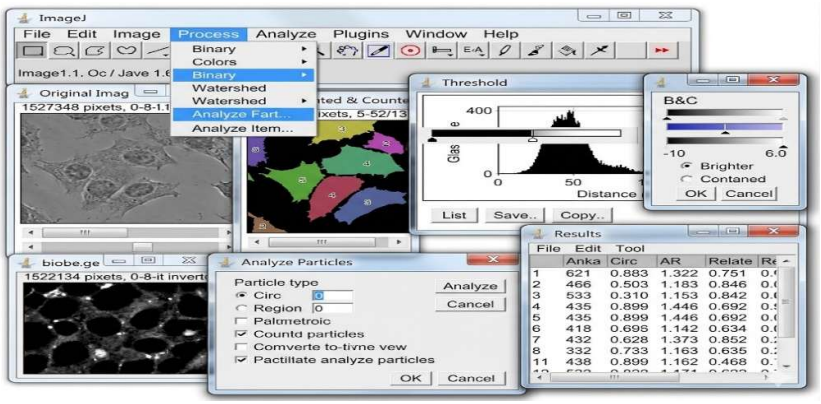


Figure 8. ImageJ

1.4 MATHEMATICAL REPRESENTATION OF AN IMAGE

In digital image processing an image is not just a visual object but a structured mathematical entity. Representing an image mathematically allows us to process, analyze, using algorithms.

1.4.1 Image as a Function

A digital image is commonly represented as a two-dimensional function:

$$f(x, y) \tag{1}$$

where:

x and y are spatial coordinates (position of a pixel in the image).

$f(x, y)$ represents the intensity (or brightness) at that point.[1]

For grayscale images, the value of f (x, y) typically ranges from 0 to L to L – 1, where L is the number of possible intensity levels (e.g.,256 for 8-bit images).

- **Continuous vs Digital Images**

In real-world scenarios, images are continuous in nature. However, computers work with discrete values. Therefore, conversion is required:

(a) Continuous Image

$$f(x, y) \tag{2}$$

x, y: continuous values

Intensity: continuous

(b) Digital Image

$$f[m, n] \tag{3}$$

m, n: discrete pixel indices

Intensity: quantized values

This conversion involves two steps:

- Sampling → converting spatial coordinates into discrete values
- Quantization → converting intensity into discrete levels

1.4.2 Matrix Representation of an Image

A digital image can be represented in matrix form:

$$f[m, n] = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0, N-1) \\ f(1,0) & f(1,1) & \dots & f(1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1, N-1) \end{bmatrix} \tag{4}$$

where:

M = number of row (height)

N = number of column (width)

Each element of the matrix corresponds to a pixel value.[1]

- **Mathematical Operation on Images**

Since image is a matrix, various mathematical operations can be applied:

- **Addition** → Image Blending

$$g(x,y) = f_1(x,y) + f_2(x,y) \tag{5}$$

- **Scalar Multiplication** → Brightness control

$$g(x,y) = k \cdot f(x,y) \tag{6}$$

- **Filtering (Convolution)** → Smoothing, Sharpening

$$g(x,y) = f(x,y) * h(x,y) \tag{7}$$

1.5 MEASUREMENT PARAMETER OF IMAGES

Measurement parameters are essential tools used to evaluate the performance of image processing and segmentation techniques. These parameters provide a quantitative way to measure the quality, accuracy, and effectiveness of the processed image compared to the original image. They help in making objective decisions instead of relying only on visual inspection.

The need for measurement parameters arises because image processing results must be validated for correctness and efficiency. These parameters help in determining how well noise has been removed, how accurately segmentation has been performed, and how closely the processed image matches the original image. They are especially important in applications such as medical imaging, where precision is very critical.[1],[5]

1.5.1 Types of Measurement Parameter

Some measurement parameter of images is

1. PSNR Ratio
2. Jaccard Index
3. Dice coefficient
4. F1 Score
5. Accuracy

1. **PSNR Ratio**

PSNR is a widely used metric to measure the quality of reconstruction of an image/video signal compared to the original. Higher PSNR values indicate better quality (less distortion).[1]

$$PSNR = 10 * \log_{10}\left[\left(\frac{MAX^2}{MSE}\right)\right] \quad (8)$$

where

MAX = maximum possible pixel value (255 for 8-bit images)

Mean Square Error between original and reconstructed images

$$MSE = \left(1/(m*n)\right) * \sum i \sum j\,[I(i,j) - k(i,j)]^2 \quad (9)$$

where

$I(i,j) =$ Original image pixel

$k(i,j) =$ Reconstructed image pixel

$m * n =$ Image dimensions

Typical value of PSNR

Table 1. Range of PSNR

Quality	PSNR Range (dB)
Excellent	> 40
Good	35-40
Fair	30-35
Poor	20-30
Very Poor	< 20

1. Jaccard coefficient

Jaccard Coefficient measures similarity between two sets by calculating the ratio of their intersection to union. Commonly used in image segmentation, object detection, and binary classification.[5]

$$Jaccard(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{TP}{(TP + FP + FN)} \quad (10)$$

where

TP = True positive

FP = False positive

FN = False Negative

Range: [0,1], (1 = Perfect overlap, 0 = No overlap)

Typical value of Jaccard Index

Table 2. Range of Jaccard index

Quality	Range
Excellent	0.75-0.80
Good	0.65-0.75
Acceptable	0.55-0.65
Poor	< 0.55

2. Dice coefficient

Dice Coefficient measures overlap between two sets (predictions vs ground truth). Most popular metric in medical image segmentation due to its balance between precision and recall.[5]

$$Dice\ coefficient = 2 * |A \cap B|/(|A|) + |B| \quad (11)$$

where

A = Ground Truth segmentation

B = Predicted segmentation

Range: [0,1], (0 = No overlap, 1 = Perfect overlap)

Typical value of Dice coefficient

Table 3. Range of Dice Coefficient

Quality	Range
Excellent	> 0.90
Good	0.80-0.85
Acceptable	0.70-0.80
Poor	< 0.70

3. F1 Score

The F1 Score is defined as the harmonic mean of precision and recall. It gives a single score that reflects both the correctness and completeness of the model's predictions.[5]

$$F1 = 2 * \frac{(Precision * Recall)}{Precision + Recall} \quad (12)$$

Where

Precision = $\frac{TP}{TP+}$

Recall = $\frac{TP}{TP+F}$

Typical value of F1 score

Table 4. Range of F1 score

Quality	Range
Excellent	> 0.90
Good	0.85-0.90
Acceptable	0.75-0.85
Poor	< 0.75

4. Accuracy

Accuracy is defined as the ratio of correctly predicted pixels (both positive and negative) to the total number of pixels in the image. It provides an overall measure of correctness of the model.[5]

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (13)$$

Typical value of Accuracy

Table 5. Range of accuracy

Quality	Range (%)
Excellent	> 95
Good	85-95
Acceptable	70-85
Poor	< 70

1.5.2 Comparison of Measurement Parameter

Table 6. Comparison table of measurement parameter

PARAMETERS	DEFINITION	FEATURES	IDEAL RANGE
Dice - coefficient	Dice - coefficient is used to measure the similarity between predicted and actual regions based on their overlap.	(a) More overlap → higher Dice value → better segmentation (b) Less overlap → lower Dice value → poor segmentation	> 0.9 preferred
PSNR	PSNR measures how close a processed image is to the original.	Higher value = better quality, less noise	> 40 dB is preferred
Jaccard Index	Jaccard index measures overlap between predicted and actual regions.	Higher Jaccard index = better prediction accuracy	≥ 0.85 → very accurate segmentation
Accuracy	Accuracy measures the percentage of correctly classified pixels in an image.	Higher = better	0.90 - 1.0
Mean Squared Error	MSE measures error between images.	Lower value = better quality	close to 0 (typically, < 50 for good quality)
Root Mean Squared Error	RMSE is the square root of MSE, used to measure the average error between original and predicted image pixels in the same units as pixel values.	Lower RMSE = better image quality	0 – 15 (good), closer to 0 is best

Structural Similarity Index - Simplified	SSIM measures the similarity between two images based on structure, brightness, and contrast, making it closer to human visual perception than PSNR or MSE.	Higher SSIM = better perceptual quality	0.95 – 1.0

CHAPTER 2

2.TECHNICAL BACKGROUND & LITERATURE REVIEW

2.1 TECHNICAL BACKGROUND

Medical image segmentation is an important process in medical imaging and healthcare applications. It is used to divide medical images into meaningful regions such as organs, tissues, tumors, and abnormal areas. Segmentation helps doctors in disease diagnosis, treatment planning, and medical analysis [6]. Modern medical imaging techniques such as MRI, CT scan, ultrasound, and X-ray generate large amounts of image data. Manual analysis of these images is difficult and time-consuming. Therefore, researchers developed automatic and semi-automatic image segmentation techniques to improve diagnostic accuracy and reduce human effort [7].

The literature review is the study of previously published research papers, journals, and articles related to medical image segmentation. It helps researchers understand existing techniques, advantages, limitations, and recent developments in this field. Through literature review, researchers can identify research gaps and improve existing segmentation methods [6]. Initially, traditional image processing methods such as thresholding, edge detection, region growing, and clustering techniques were used for segmentation. These methods are simple and computationally efficient but may not provide accurate results for noisy and complex medical images [7].

Later, machine learning and deep learning techniques were introduced to improve segmentation performance. Deep learning models such as Convolutional Neural Networks (CNN) and U-Net architectures achieved high accuracy in medical image analysis and disease detection [8]. Recent research focuses on advanced deep learning and attention-based models that provide better segmentation performance for MRI, CT, ultrasound, and X-ray images. These techniques support computer-aided diagnosis systems and improve healthcare applications [6]. These modern techniques reduce human effort, improve diagnostic accuracy, and support computer-aided diagnosis systems in hospitals and healthcare centers. The literature review presented in this chapter focuses on different medical image segmentation techniques, their applications in healthcare, evaluation methods, advantages, limitations, and recent developments in deep learning-based segmentation systems. This review provides a strong theoretical foundation for understanding the proposed project work.

2.2 LITERATURE SURVEY

Several researchers have proposed different medical image segmentation techniques to improve segmentation accuracy and computational efficiency. The summary of important research works is presented in Table 7.

Table 7. Literature survey

S.NO.	Author & Year	Technique Used	Key Findings	Limitations
01	Hesamian et al. (2019)	Deep Learning-based Segmentation	Deep learning methods improved segmentation accuracy in medical imaging applications.	Requires large annotated datasets and high computational cost.
02	Dar and Padha (2019)	Thresholding, Watershed, ANN	Reviewed traditional segmentation techniques and their applications in medical imaging.	Traditional methods are sensitive to noise and intensity variation.
03	Mohapatra et al. (2024)	K-means, Region Growing, Graph Cut	Conventional segmentation methods are clinically useful for liver image analysis.	Lack of common datasets makes comparison difficult.
04	Mittal et al. (2021)	Clustering-based Segmentation	K-means clustering is computationally efficient and widely used in image segmentation.	Sensitive to initial centroid selection and local minima problem.
05	Wang et al. (2020)	Deep Learning using CNN and U-Net	U-Net architecture achieved better performance in medical image segmentation.	Requires powerful hardware and long training time.
06	Sumithra and Malathi (2018)	Thresholding and Clustering Method	Segmentation techniques are effective for extracting regions of interest in medical images.	Accuracy decreases for low-contrast images.
07	Jena et al. (2018)	Machine Learning-based Segmentation	Machine learning techniques improved automated medical image analysis.	Performance depends on training data quality.
08	Armya and Abdulazeez (2020)	Unsupervised Algorithms (K-means, PAM)	Unsupervised methods reduce manual intervention in segmentation.	Difficult to achieve high accuracy in complex images.
09	Litjens et al. (2017)	Deep Learning in Medical Imaging	Deep learning significantly improved medical image analysis performance.	Requires large datasets and GPU-based computation.
10	Zawish et al. (2019)	Brain Tumor Segmentation	Region-based and clustering methods are useful for tumor detection in MRI images.	Segmentation performance affected by image noise and artifacts.

Recent deep learning-based segmentation methods such as CNN and U-Net architectures have significantly improved segmentation accuracy. However, these methods require large annotated datasets, powerful GPU hardware, and long training times [3], [4]. In medical imaging applications, obtaining labeled datasets is difficult because annotation requires expert radiologists and considerable time [5], [6].

Researchers have also identified the issue of domain variation in medical images. Images obtained from different hospitals, imaging devices, and scanning protocols often vary in quality and intensity, reducing the generalization capability of segmentation models [8].

Furthermore, many existing methods focus mainly on improving segmentation accuracy but ignore computational efficiency and real-time implementation. This creates difficulties in deploying segmentation systems in low-resource healthcare environments [1], [4].

Therefore, there is a need to develop an efficient medical image segmentation technique that:

i. Provides accurate segmentation results for noisy and low-contrast images.

ii. Requires less computational complexity and memory usage.

iii. Works effectively with limited training data.

iv. Reduces dependency on expensive hardware resources.

v. Improves interpretability and reliability for clinical applications.

vi. Enhances segmentation accuracy while maintaining fast processing time.

The proposed work aims to address these limitations by developing an improved image segmentation approach using efficient preprocessing and clustering techniques for medical image analysis.

2.4 TRADITIONAL MEDICAL IMAGE SEGMENTATION TECHNIQUES

Traditional medical image segmentation techniques are the early methods used to divide medical images into meaningful regions such as organs, tissues, tumors, and blood vessels. These methods are mainly based on image processing algorithms and mathematical operations. Traditional techniques are simple, computationally efficient, and easy to implement. They are widely used in MRI, CT scan, ultrasound, and X-ray image analysis [6].

The main objective of traditional segmentation methods is to identify the boundaries and structures present in medical images. However, medical images often contain noise, low contrast, intensity variation, and overlapping tissues, which reduce segmentation accuracy. Despite these limitations, traditional methods are still important because they form the foundation for modern machine learning and deep learning techniques [7].

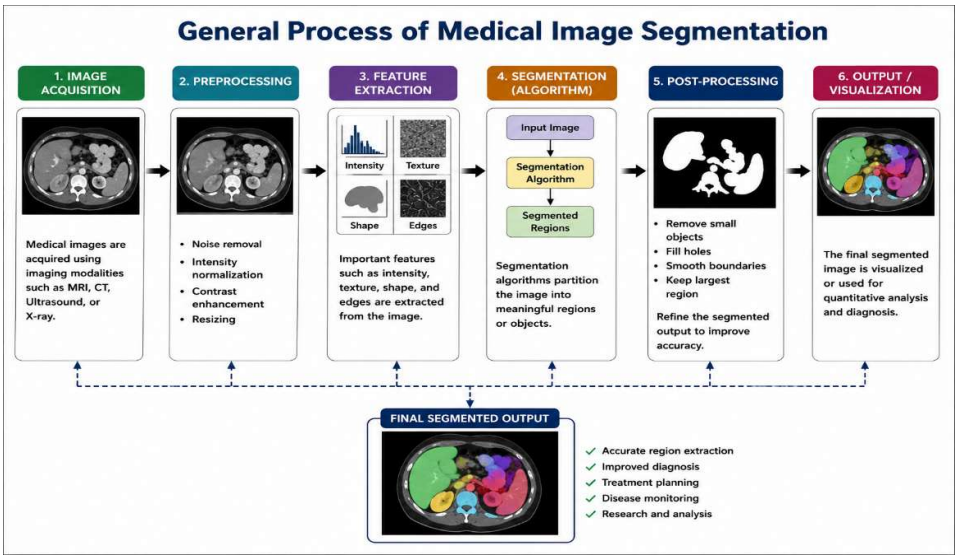


Figure 9. General Process of Medical Image Segmentation

The commonly used traditional medical image segmentation techniques are explained below.

1. Threshold-Based Segmentation.
2. Edge-Based Segmentation.
3. Region-Based Segmentation.
4. Clustering-Based Segmentation.
5. Watershed Segmentation.

2.5 MACHINE LEARNING-BASED SEGMENTATION TECHNIQUES

Machine Learning (ML)-based segmentation techniques are advanced methods used to improve the accuracy and automation of medical image analysis. Unlike traditional image processing methods, machine learning algorithms can learn important features from medical images and classify different tissues, organs, and abnormal regions automatically. These techniques are widely used in MRI, CT scan, ultrasound, and X-ray image segmentation for disease diagnosis and healthcare applications [7].

Machine learning methods reduce human effort and improve segmentation performance by analyzing large medical datasets. These techniques generally involve two important stages:

1. **Feature Extraction** – Important image features such as texture, shape, intensity, and edges are extracted.
2. **Classification** – Extracted features are classified into different categories such as normal tissue, tumor, or abnormal region [6].

2.5.1 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and segmentation tasks in medical image processing. SVM classifies image pixels into different categories by creating an optimal decision boundary called a hyperplane.

This hyperplane separates different classes with maximum margin, which improves classification accuracy [7].

SVM is widely used in various medical imaging applications, including:

i. Brain tumor segmentation in MRI images
ii. Lung disease diagnosis in CT scans
iii. Breast cancer detection
iv. Liver and kidney segmentation
v. Tissue classification in ultrasound images

2.5.2 Artificial Neural Network (ANN)

Artificial Neural Network (ANN) is a machine learning technique inspired by the structure and working of the human brain. ANN consists of interconnected processing units called neurons, which work together to process and analyze image data. ANN learns patterns from training datasets and performs classification and segmentation tasks automatically [7].

An ANN generally consists of three layers:

1. **Input Layer** – Receives image data and extracted features.
2. **Hidden Layer** – Processes information using interconnected neurons.
3. **Output Layer** – Produces segmentation or classification results.

ANN-based segmentation techniques are widely used in:

i. Brain tumor detection in MRI images
ii. Lung disease analysis in CT scans
iii. Breast cancer detection
iv. Organ and tissue segmentation
v. Pattern recognition in medical imaging

2.5.3 Random Forest Algorithm

Random Forest is a supervised machine learning algorithm used for classification, prediction, and segmentation tasks in medical image processing. It is an ensemble learning method that combines multiple decision trees to improve segmentation accuracy and reduce errors. Each decision tree independently analyzes image features, and the final result is determined by majority voting among all trees [6].

Random Forest is widely used in various healthcare applications, including:

i. Brain MRI segmentation
ii. Tumor detection
iii. Organ and tissue classification
iv. Ultrasound image analysis
v. Lung disease detection in CT scans

CHAPTER 3

3. MEDICAL IMAGE SEGMENTATION TECHNIQUES AND K-MEANS CLUSTERING

3.1 MEDICAL IMAGE SEGMENTATION

Medical Image Segmentation is an important technique in digital image processing used to divide a medical image into meaningful regions such as organs, tissues, or abnormalities. It helps doctors to analyse medical images more accurately and efficiently. Segmentation plays a vital role in identifying diseases like tumours, infections, and other abnormalities in images obtained from MRI, CT scan, and X-ray.

Medical image segmentation refers to the process of partitioning a digital image into multiple region (sets of pixel). The goal of segmentation is to change the representation of image into something that is more meaningful and easier to analyse. More precisely, image segmentation is the process of grouping an image into units or categories that are homogeneous with respect to one or more characteristics. The result of image segmentation is a set of segments that collectively cover the entire image, or a set of contours extracted from the image. The pixels in the same region have similar gray scale or spectral values or belong to a similar pattern and form a connected region. Adjacent regions are significantly different with respect to the same characteristics.

A formal definition [Pal *et al.* 1993] of image segmentation is as follows: If P is a homogeneity predicate defined on groups of connected pixels, then segmentation is a partition of the set I into connected subsets or regions (S1, S2,S3,……, Sn) such that U Si=1,2,3,……,n = I with Si u Sj = Ø (I U j). The uniformity predicate P(Si) is true for all regions Si, and P (Si U Sj) is false, when I u j and Si and Sj are neighbors. Segmentation is used for many applications, for example, medical diagnosis like, CT scan, MRI scan image compression, mapping and measurement. Segmentation techniques are used to identify and extract features from remotely sensed imagery.[1]

3.2 SEGMENTATION TECHNIQUES

Segmentation techniques could be divided into spectral and spatial segmentation. Spectral segmentation method assigns a pixel to a region according to the spectral similarities. It relies on purely spectral features (spectral reflectance characteristics) and considers only one pixel at a time. Spatial segmentation method categorizes an image pixel on the basis of their spatial relationship with the pixels surrounding them. It may consider aspects such as image texture, feature size, directionality, context and repetition. Thresholding, clustering, neural networks segmentation, fuzz segmentation and edge-based segmentation belong to the spectral segmentation, whereas region growing, texture-based segmentation and mathematical morphology, splitting and merging and object-based approach belong to the spatial Segmentation.[1]

3.2.1 Thresholding

Thresholding is perhaps the most common segmentation technique and is the most basic region-segmentation technique. The technique separates pixels into background and foreground (object of interest) classes based upon their similarities in gray-level intensity. To implement this technique, a threshold (T) value is chosen. Every pixel in the image is then compared to the T value. Each pixel point (x, y) is given a region label of "0" (background) if $f(x, y) \leq T$ i.e. the pixel value of point (x, y) is less than or equal to T or "1" (foreground) if $f(x, y) > T$ i.e. the pixel value of point (x, y) greater than T. This form of region segmentation results in a binary image, in which each region is either white (1) or black (0). [10]

The threshold value is critical, as it sets the discriminating criterion for the segmentation process.

$$g(x, y) = \begin{cases} 1 & if\ f(x, y) > T \\ 0 & if\ f(x, y) \leq T \end{cases} \quad (14)$$

where

$f(x, y)$ = original image

$g(x, y)$ = segmented image

T = threshold value

3.2.2 Edge-Based Segmentation

In edge-based segmentation approach, the aim is to find edges between the regions and determine the segments as regions within these edges. From this point of view, edges are regarded as boundaries between image objects and they are located where changes in values occur. There are various ways to delineate boundaries but in general the first step of any edge-based segmentation approach is edge detection which consists of three steps [Jain *et al*. 1995]: filtering, enhancement and detection. Filtering step is usually necessary in decreasing the noise present in the imagery. The enhancement aims to the revealing of the local changes in intensities. One possibility to implement the enhancement step is to use high-pass filtering. The actual edges are detected from the enhanced data using thresholding technique. Finally, the detected edge points are linked to form the region boundaries. This technique is extensively utilized for gray level image segmentation, which is based on the detection of discontinuity on gray level, and tries to locate points with abrupt changes in gray level. In contrast to cluster-based and most region-based segmentations, it cannot be performed in parallel on all pixels. The edge-based segmentation takes into account of an object that is characterized by adjacent pixels and describes the segments by their outlines.[1]

3.2.3 Region-Based Segmentation

This technique attempts to group pixels into homogeneous regions with similar properties. The ideas of region-based algorithms come from the observation that pixels inside a structure tend to have similar intensities. Region growing algorithm is a typical algorithm of this type. It groups method is although simple in nature but there are three problems in this method:

i. Selection of initial seeds that properly represent region of interest.
ii. Selection of suitable properties for including points in the various regions during the process.
iii. The formulation of stopping rule.

The criteria such as intensity, texture and color are used to grow the region. The additional criteria that increase the power of region growing are size, likeness between candidate pixels and the pixels grown so far (such as a comparison of intensity of a candidate and the average intensity of the region), and the shape of the region being grown. Growing a region is stopped when no more pixels satisfy the criteria for inclusion in that region.[1]

3.2.4 Clustering-Based Segmentation

Clustering is the organization of collection of patterns into a group. Clustering algorithm is often used to perform image segmentation. In this approach pixels are mapped into various feature spaces whereupon they are subjected to grouping algorithm. K-mean [Hartigan *et al.* 1979] is a popular clustering method used for segmenting the image. It is an iterative technique that is used to partition an image into K-clusters.

The basic algorithm is:

1. Pick K cluster centres either randomly or based on some heuristic.
2. Assign each pixel in the image to the cluster that minimize the distance between the pixel and the cluster center.
3. Re-compute the cluster centers by averaging all the pixels in the cluster.
4. Repeat steps 2 and 3 until convergence is attained.

In this case, distance is the squared or absolute difference between a pixel and a cluster centre. The difference is typically based on pixel intensity. This algorithm is guaranteed to converge but it may not return to the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K. Typically, K-means is run independently for different values of K and the partition that appears the most meaningful to the domain expert is selected. Different initializations can lead to different final clustering because K-means only converges to local minima. One way to overcome the local minima is to run the K-means algorithm, for a given K, with multiple different initial partitions and choose the partition with the smallest squared error.[11]

3.2.5 Watershed Segmentation

The watershed algorithm divides an image into segments using topographic information. It treats the image as a topographic surface, identifying catchment basins based on pixel intensity.

Local minima are marked as starting points, and flooding with colors fills catchment basins until object boundaries are reached. The resulting segmentation assigns unique colors to regions, aiding object recognition and image analysis.

The whole process of the watershed algorithm can be summarized in the following steps:

i. **Marker placement:** The first step is to place markers on the local minima, or the lowest points, in the image. These markers serve as the starting points for the flooding process.
ii. **Flooding:** The algorithm then floods the image with different colors, starting from the markers. As the color spreads, it fills up the catchment basins until it reaches the boundaries of the objects or regions in the image.
iii. **Catchment basin formation:** As the color spreads, the catchment basins are gradually filled, creating a segmentation of the image. The resulting segments or regions are assigned unique colors, which can then be used to identify different objects or features in the image.
iv. **Boundary identification:** The watershed algorithm uses the boundaries between the different colored regions to identify the objects or regions in the image. The resulting segmentation can be used for object recognition, image analysis, and feature extraction tasks.[1]



Figure 10. Segmentation technique

3.3 OTSU THRESHOLDING

Otsu method's is a global thresholding technique that automatically performs image thresholding. It finds the optimal threshold to separate foreground objects from the background by minimizing intra-class variance. The method is widely used in image segmentation and is effective for image with bimodal histograms. It operates by maximizing the between-class variance to determine the optimal threshold value that the best separates pixel intensities into foreground and background classes. The method is simple and effective, making it a popular choice for automatic image thresholding and segmentation.[10]

The between-class variance σ^2 b(t) for threshold t is defined as:

$$\sigma^2 b(t) = \omega_0(t)\omega_1(t)[[\mu_0(t) - \mu_1(t)]^2] \quad (15)$$

3.3.1 Otsu Algorithm

Otsu thresholding was proposed by Otsu [30] in 1979 for image thresholding where a criterion function generates some sort of measure of dissociation between regions of the image. Otsu thresholding selects the threshold value that minimizes intra-class variation of the foreground and background pixels. Lists the steps of the Otsu thresholding algorithm as shown in:[10]

Step 1: Convert the image into grayscale.

Step 2: Compute the grayscale Histogram.

Step 3: Compute the probability of each intensity level.

Step 4: Calculate the class means and class variances.

Step 5: Compute between-class variance.

Step 6: Select the threshold value that maximizes the between-class variance.

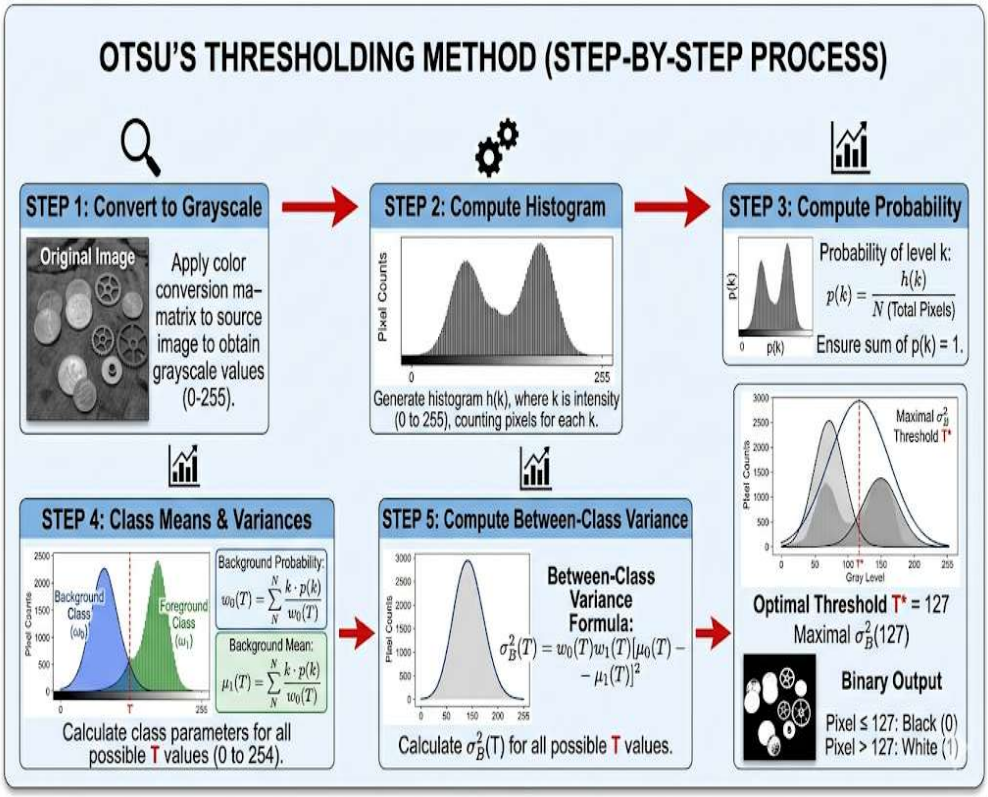


Figure 11. Step of OTSU Method

3.3.2 Mathematical foundation of OTSU Thresholding

Consider a grayscale image I : Ω → {0,1,...,L − 1} where Ω represents the spatial domain and L = 256 for 8-bit imagery. Let h(i) denote the histogram frequency of intensity level i, with total pixel count given by:[10]

$$N = \sum_{i=0}^{L-1} h(i) \tag{16}$$

The normalized probability mass function is defined as:

$$p(i) = \frac{h(i)}{N} \tag{17}$$

For a candidate threshold t ∈ [0,L −1], the image is partitioned into two classes: C0 = {0,1,...,t − 1} (background) and C1 = {t,t+1,...,L−1} (foreground). The class probabilities are computed as:

$$\omega_0(t) = \sum_{i=0}^{t=1} p(i) \tag{18}$$

$$\omega_1(t) = \sum_{i=t}^{L=1} p(i) \tag{19}$$

The class means are calculated as:

$$\mu_0(t) = \frac{\sum_{i=0}^{t-1} i \cdot p(i)}{\omega_0(t)} \tag{20}$$

$$\mu_1(t) = \frac{\sum_{i=t}^{L-1} i \cdot p(i)}{\omega_1(t)} \tag{21}$$

The OTSU criterion maximizes the between-class variance:

$$\sigma_B^2(t) = \omega_0(t) \cdot \omega_1(t)[\mu_1(t) - \mu_0(t)]^2 \tag{22}$$

The optimal threshold is determined by:

$$t^* = \arg \max_{t \in [0, L-1]} \sigma_B^2(t) \tag{23}$$

3.4 MODIFIED OTSU THRESHOLDING TECHNIQUE

Otsu's method of image segmentation is one of the best methods for threshold selection. With Otsu's method an optimum threshold is found by maximizing the between-class variance; Otsu algorithm is based on the gray-level histogram which is estimated by a sum of Gaussian distributions. In some type of images, image data does not best fit in a Gaussian distribution model. The objective of this study is to develop and compare modified versions of Otsu method, one is based on **Log-normal distribution (Otsu-Lognormal),** while the other is based on **Gamma distribution (Otsu-Gamma), Weighted object variance method** the maximum between-cluster variance is modified based on each model. The two proposed methods were applied on several images and promising experimental results were obtained. Evaluation of the resulting segmented images shows that both Otsu-Gamma method and Otsu-Lognormal yield

better estimation of the optimal threshold than does the original Otsu method with Gaussian distribution (Otsu).[12]

3.4.1 Weighted Object Variance (WOV) Method

In the OTSU method, the defect-free image segmentation is neglected. Most thresholding methods can correctly separate defects from their background, but wrongly conclude a defect-free image into the detective image. The object area is normally much smaller than the background area. Therefore, the desired threshold should be close to a low grey level compared with the threshold generated by the OTSU threshold. The OTSU threshold is given:

$$T_h = \frac{(u_0(t) + u_1(t)}{2} \tag{24}$$

This OTSU threshold makes the object and background have a similar size. The desired threshold should keep the probability of the object at a small value or even equal to 0. To get the desired threshold, the first term i.e. the object variance of between-class variance shown in below should contribute less to the and the thresh- old should be mainly decided by the second term of between-class variance i.e. the background variance.

$$Y_b(t) = wP_0(t)\big(u_0(t)\big)^2 + P_1(t)(u_1(t))^2 \tag{25}$$

The first term in between-class variance, given in above is defined as the object variance and the second term is the background variance. A parameter ω ranging from 0 to 1 is weighted with first term and keeping the second term unchanged. The weighted object variance is ω times the original object variance.

A weighted parameter equal to the cumulative probability of the defect occurrence is a better selection for defect detection. To avoid false detection, the object variance should contribute less. The problem with using this method is that the defects are very small as compared to the background. This method is valid when the object is small compared to the background, but when the object and background are similar or bigger than that of the background, this method fails to give the desired threshold. Generally, the main criteria are converting an image into a binary image and selecting the best threshold. The OTSU method is based on a pixel intensity distribution histogram. The color images contain a lot of information. Three matrices are allocated for the colored images named red channel, green channel, and blue channel, which require a high amount of computational power and computation time. To reduce the computational time, the first image is converted into the grey level. The objects can be detected according to their grey-level information.

Although, it is tricky to select an optimal threshold to separate an object from the image background because of the following factors:

1. Because of illumination inequalities and low contrast between the object and background.

2. Sometimes the object is much smaller than that of the background and vice-versa, it is difficult to locate the object as the size of the object in the frame is not the same. Thus, the image histogram is not always bimodal.
3. Using the 2D or 3D OTSU method may not work. As the complexity increase the computation time also increases, so it cannot be used in real-time application.

By scrutinizing the problems in the existing thresholding methods for object detection, we formulate an Improved Thresholding method for object detection. The method is an improved version of Gaussian OTSU and Weighted object variance OTSU. The threshold value obtained by the proposed OTSU method gives better results and an optimal threshold value.[12]

3.4.2 The Log-Normal Distribution

Most of the images in nature are asymmetric, in particular in color images, the perceived object wavelength spectrum may not be symmetric. Hence it is needed to consider asymmetric distributions for effective analysis of images. Many models have been highlighted in the literature for catering the asymmetric nature of distributions. Among these distributions skew, Log-normal, Gamma is mostly highlighted in the literature. Among the various asymmetric models Log normal distribution is considered for the experimentation in this paper. The main advantage of considering log normal distribution is that it caters the images which are both symmetric and non-symmetric in nature.

In probability theory, a log-normal distribution is a probability distribution of a random variable whose logarithm is normally distributed. If Y is a random variable with a normal distribution, then $X = exp\ (Y)$ has a log-normal distribution; likewise, if X is log-normally distributed, then $Y = \log(X)$ is normally distributed. In the case of a digital image whose histogram $h(g)$ is assumed to be modeled as log-normal, then it can be defined as:[12]

$$h(g) = \sum_{i=1}^{m} pi \cdot f(x, \mu_i, \sigma_i), g > 0 \tag{26}$$

where $(pi \cdot f(x, \mu_i, \sigma_i))$ is the ith mode in the histogram, pi is the probability of this i^{th} mode, and m is the number of modes in that histogram. The probability density function (pdf) of the log-normal distribution is defined as:[12]

$$F(g, \mu, \sigma) = \frac{1}{g\sigma\sqrt{2\pi}} \cdot e^{\frac{1}{2}\left(\frac{lng - \mu}{\sigma}\right)^2} \tag{27}$$

where

g: pixel's intensity level (e.g. grey level)

σ: standard deviation

μ: mean;

Now, if the data in an image is assumed to be modeled by a Log-normal distribution, then, one can see the image *I*(*x, y*) is composed by two Log-normal distributions. Therefore, u_{0Log} and u_{1Lo} can be estimated as follows:[12]

$$\mu_{0L}^2 \quad (t) = \frac{\sum_{i=0}^{t-1} h(i) \cdot \log(i)}{\sum_{i=1}^{t-1} h(i)} \tag{28}$$

$$\mu_{1Lo}^2 \ (t) = \frac{\sum_{i=t}^{L} h(i) \cdot \log(i)}{\sum_{i=1}^{L} h(i)} \tag{29}$$

3.4.3 The Gamma Distribution

Gamma distribution is a continuous probability distribution used to model **skewed intensity values** in an image. In image segmentation, it helps in separating objects from the background when the intensity histogram is not symmetric (i.e., not normally distributed). Unlike simple thresholding methods, gamma distribution provides a **statistical approach** to model pixel intensities more accurately.

The gamma distribution, like the lognormal distribution, is an alternative to consider for data that seem to be highly skewed. Gamma distribution is more general than Gaussian and it can be encountering the limitations of Gaussian due to its ability to provide symmetric and non-symmetric histograms. For a Gamma distribution with parameters α and β, the probability density functions in homogeneous area is given by:[12]

$$F(g, \mu, N) = \frac{2q}{\mu} \cdot \frac{N^N}{\Gamma(N)} \cdot \left(\frac{qx}{\mu}\right)^{2N-1} \cdot e^{-N\left(\frac{qx}{\mu}\right)^2} \tag{30}$$

Where q = $\frac{\gamma(N+0.5)}{\sqrt{N}.\gamma(N)}$, x is the intensity of the pixel, μ is the mean value of the distribution and N is the shape of distribution. The shape of the Gamma distribution could be symmetry or skewed to the right. If the data in an image is assumed to be modeled by a Gamma distribution, then, one can see the image *I*(*x, y*) is composed by two Gamma distributions.

Therefore, μ_{0Gamma} and μ_{1Gamma} can be estimated as follows:[12]

$$\mu_{0Gamma}^2(t) = \frac{\sum_{i=1}^{t-1} h(i) \cdot i^2 \cdot q^2}{\sum_{i=1}^{t-1} h(i)} \tag{31}$$

$$\mu_{1Gamma}^2(t) = \frac{\sum_{i=t}^{L} h(i) \cdot i^2 \cdot q^2}{\sum_{i=t}^{L} h(i)} \tag{32}$$

Between-class variance:

$$\sigma_B^2(t) = \frac{[\mu_T \omega_0 - \mu(t)]^2}{\omega_0[1 - \omega_0]} \tag{33}$$

Where

μ_t: total mean

ω_0: probability of class 0

3.5 K-MEANS CLUSTERING METHOD

K-Means clustering is an unsupervised machine learning algorithm used for grouping similar data points into clusters. In image segmentation, each pixel is treated as a data point, and pixels with similar characteristics are grouped into the same cluster [19], [20].

The main objective of K-Means clustering is to minimize intra-cluster variation and maximize inter-cluster variation [18].

The algorithm works through the following major steps:

1. Selection of number of clusters (K)
2. Initialization of cluster centroids
3. Distance calculation between pixels and centroids
4. Assignment of pixels to nearest clusters
5. Recalculation of cluster centroids
6. Repetition of the process until convergence

Mathematical Representation of K-Means

Suppose the dataset is represented as:

$$X = \{x_1, x_2, x_3, \ldots\ x_n\} \tag{34}$$

where:

x_i represents image pixels or feature vectors

n represents the total number of pixels

The K-Means algorithm divides the dataset into K clusters.

The objective function minimized by K-Means is [18]:

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \left\| x_i - c_j \right\|^2 \tag{35}$$

where:

$J =$ objective function

$x_i =$ pixel value

$c_j =$ pixel value

The distance between pixel and centroid is calculated as [18]:

$$d = \sqrt{\left(x_i - c_j\right)^2} \tag{36}$$

The centroid of each cluster is updated using [20]:

$$C_j = \frac{1}{N_j} \sum_{i=1}^{N_j} x_i \tag{37}$$

where:

C_j = new centroid

N_j = number of pixels in cluster

CHAPTER 4

4.RESULT AND DISCUSSION

4.1 RESULT

This chapter presents the experimental results and discussion of various medical image segmentation techniques used for diagnostic applications. The performance of traditional segmentation methods and deep learning-based techniques is analyzed using medical images such as MRI and CT scans. The obtained results are evaluated using different performance metrics including Accuracy, Dice Similarity Coefficient (DSC), Jaccard Index, Precision, and Recall [1]. The objective of this chapter is to compare segmentation techniques and identify the most suitable approach for accurate medical diagnosis.

The experimental analysis includes traditional image segmentation techniques and deep learning-based methods. Traditional methods such as thresholding, edge detection, region growing, and clustering are simple and computationally efficient, but their performance decreases in noisy and complex medical images.

Before segmentation, preprocessing operations such as noise filtering and contrast enhancement are performed to improve image quality. Noise removal helps eliminate unwanted disturbances from medical images, while contrast enhancement improves the visibility of tissues and abnormal regions. These preprocessing steps significantly improve the accuracy of segmentation algorithms. The segmented images obtained from different methods are evaluated using standard performance metrics including Accuracy, Dice Similarity Coefficient (DSC), Jaccard Index, Precision, and Recall. These metrics help measure the similarity between the segmented output and the ground truth image.[3]

4.2 COMPUETR AIDED DIAGNOSTIC SYSTEM

The developed Computer Aided Diagnostic System for Preliminary Medical Image Screening provides a user-friendly graphical user interface (GUI) for performing medical image segmentation and analysis. The interface is designed using MATLAB to simplify the process of loading medical images, applying segmentation algorithms, visualizing results, and evaluating performance metrics. The GUI integrates different modules such as image upload, preprocessing, segmentation, histogram analysis, and performance evaluation into a single platform.

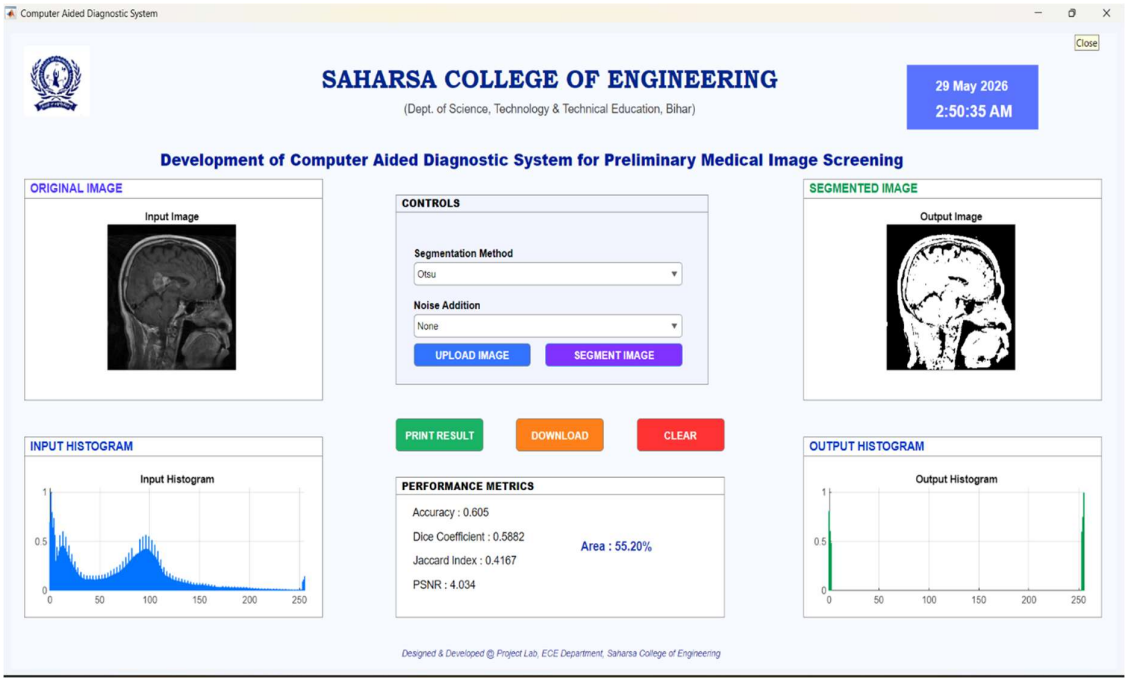


Figure 12. User interface 1

The developed Computer Aided Diagnostic System for Preliminary Medical Image Screening provides an interactive and user-friendly graphical user interface (GUI) for performing medical image segmentation and analysis. The interface is designed to simplify the process of loading medical images, selecting segmentation techniques, visualizing segmented outputs, and evaluating performance parameters. The GUI integrates different functional modules such as image upload, preprocessing, segmentation, histogram analysis, and performance evaluation into a single platform, making the system efficient and easy to operate for medical image analysis applications.

The top section of the interface contains the title of the project along with the institution name, logo, current date, and time display. This section provides proper identification of the system and improves the professional appearance of the application. A close button is also provided to safely exit the program. The central title, “Development of Computer Aided Diagnostic System for Preliminary Medical Image Screening,” clearly represents the purpose of the developed system.

On the left side of the interface, the Original Image panel is provided to display the input medical image selected by the user. This panel allows the user to visualize the original MRI, CT, or X-ray image before processing. The uploaded image acts as the primary input for segmentation and analysis. Below the original image panel, the Input Histogram section displays the histogram of the input image. The histogram represents the distribution of pixel intensity values and helps analyze image contrast and brightness. It also assists in understanding the quality of the original image before preprocessing and segmentation.

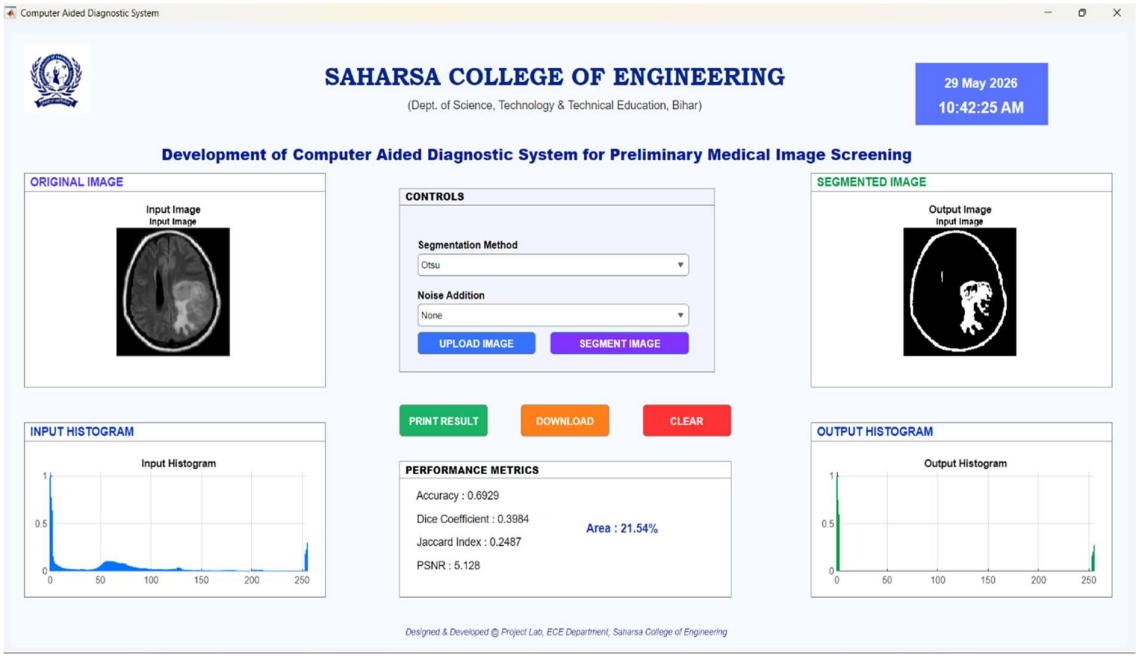


Figure 13. User interface 2

The center section of the GUI contains the Controls Panel, which manages all processing operations. The segmentation method selection option allows the user to choose different segmentation techniques such as Otsu Thresholding, K-Means Clustering, Edge Detection, and Region-Based Segmentation. In the displayed interface, the Otsu segmentation method is selected. Another option is provided for noise addition, which helps evaluate the robustness of segmentation algorithms under noisy image conditions. The Upload Image button is used to load medical images into the system, while the Segment Image button performs the segmentation operation based on the selected algorithm. These controls provide flexibility and make the system interactive and user-friendly.

The right side of the interface contains the Segmented Image panel, where the processed output image is displayed after segmentation. This section highlights the affected or abnormal region extracted from the medical image. The segmented output helps separate the region of interest from the background and improves the visualization of tumors, infected tissues, or abnormal structures. Below the segmented image panel, the Output Histogram section displays the histogram of the segmented image. This histogram helps analyze changes in pixel intensity distribution after segmentation and evaluates the effectiveness of image processing techniques.

The lower middle section of the interface contains the Performance Metrics Panel, which evaluates the efficiency and accuracy of the segmentation algorithm. Different quantitative performance parameters such as Accuracy, Dice Coefficient, Jaccard Index, PSNR (Peak Signal-to-Noise Ratio), and Affected Area are displayed in this section. Accuracy measures the correctness of segmentation results, while the Dice Coefficient and Jaccard Index measure the similarity between the segmented output and the ground truth image. PSNR evaluates the quality of the processed image, and the affected area parameter estimates the percentage of the

detected abnormal region. These metrics help compare different segmentation techniques and determine the best-performing algorithm for medical image analysis.

Additional functional buttons such as Print Result, Download, and Clear are also provided in the interface. The Print Result button is used to print segmentation outputs and analysis reports, while the Download button allows the user to save segmented images and result data. The Clear button resets the interface for processing new medical images. These additional features improve usability and make the developed system more practical for real-time applications.

The experimental results obtained from the developed GUI demonstrate that the proposed system successfully performs medical image segmentation and analysis with improved visualization and performance evaluation. Preprocessing operations enhance image quality, and segmentation algorithms accurately identify abnormal regions in medical images. The system provides real-time visualization, quantitative analysis, and efficient diagnostic support, making it suitable for preliminary medical image screening and computer-aided diagnosis applications.

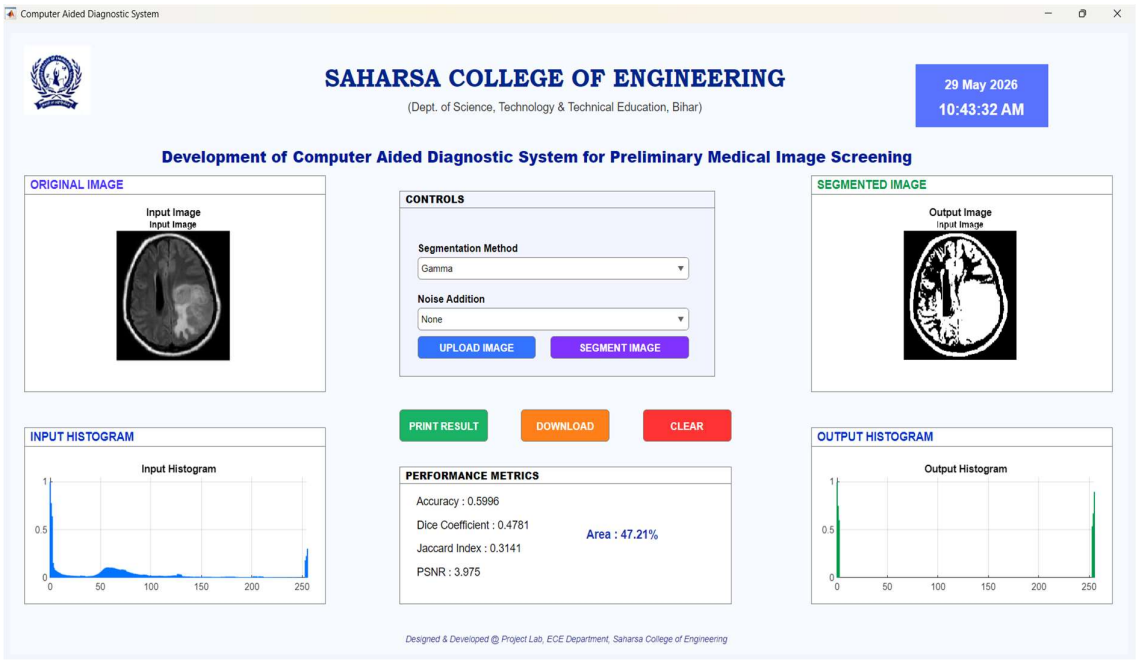


Figure 14. user interface 3

4.3 EXPERIMENTAL SETUP

The experimental setup is an important part of the medical image segmentation system because it defines the complete procedure used for image processing, segmentation, and performance evaluation. In this project, different segmentation techniques are implemented and tested on medical images such as MRI and CT scans to analyze their effectiveness for diagnostic applications [13].

The experimental setup mainly consists of the following stages:

1. Image Acquisition
2. Image Preprocessing
3. Image Segmentation
4. Feature Extraction
5. Performance Evaluation

These stages are arranged sequentially to obtain accurate segmented outputs from medical image.

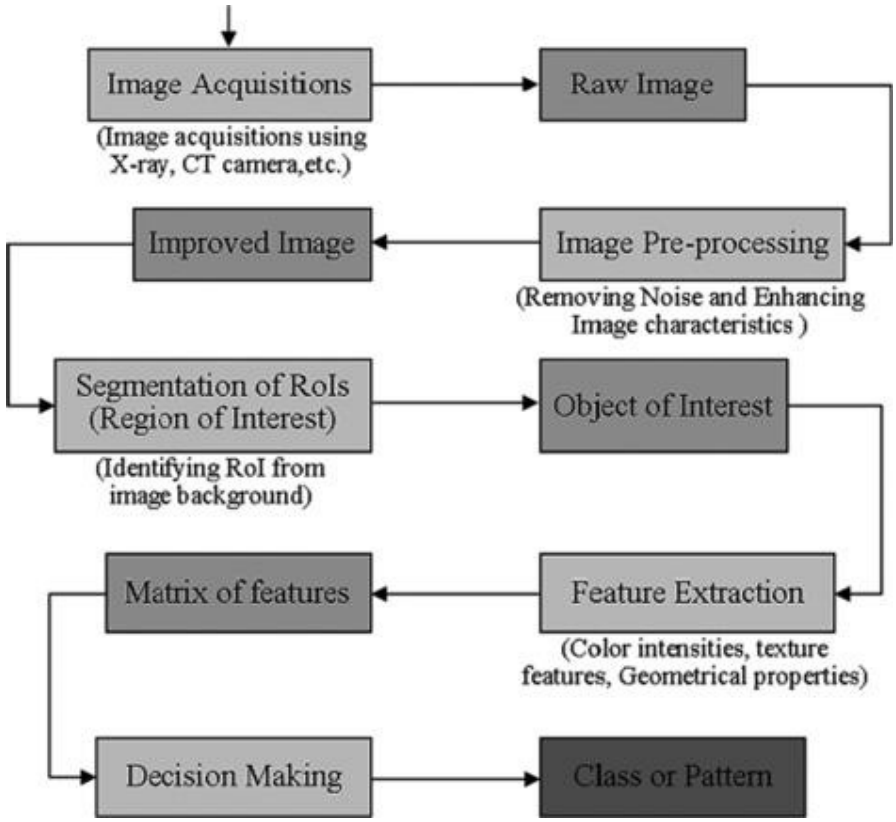


Figure 15. Technique for image processing

4.3.1 Hardware Requirement

The segmentation algorithms require a computing system with sufficient processing capability because medical images are generally large in size and computationally complex.

The hardware configuration used in this project is shown below.

Table 8. Hardware requirements used for medical image segmentation

S. No.	HARDWARE COMPONENT	SPECIFICATION
1.	Processor	Intel Core i5/i7
2.	RAM	8 GB or higher

3.	Storage	500 GB HDD/SSD
4.	Graphics Card	Integrated/Discrete GPU
5.	Operating System	Windows 10/Linux

4.3.2 Software Requirement

Software tools are required for implementing segmentation algorithms, image preprocessing, and performance analysis.

Table 9. Software tools used in experimental implementation

S.NO.	SOFTWARE TOOL	PURPOSE
1.	MATLAB	Image processing and segmentation
2.	Image Processing Toolbox	Filtering and enhancement
3.	Python\OpenCV	Additional Image analysis
4.	Medical Dataset	Input medical image

4.3.3 Dataset Description

The dataset is one of the most important components of a medical image segmentation system because the performance of segmentation algorithms depends heavily on the quality, size, and diversity of medical images used during experimentation. In this project, different medical image datasets such as MRI, CT scan, and ultrasound images are used to evaluate the effectiveness of segmentation techniques for diagnostic applications [13].

The datasets contain images of various organs and abnormal tissues including:

i. Brain tumors

ii. Lung infections

iii. Breast cancer tissues

iv. Cardiac structures

v. Retinal abnormalities

These images are collected from publicly available medical imaging databases and research repositories. The dataset is used to test the segmentation algorithms under different conditions such as noise, intensity variation, low contrast, and complex tissue structures.

4.3.4 Popular Medical Image Segmentation Dataset Resources

The following datasets are widely used in medical image segmentation research and are suitable for your final year project. These datasets contain MRI, CT, ultrasound, and tumor images with segmentation labels for training and testing segmentation algorithms.

Table 10. Dataset table

S. NO.	DATASET NAME	IMAGING MODALITIES / DATA TYPE	DATASET LINK
01.	BraTS	T1 MRI,T2 MRI, FLAIR MRI, T1-Gd MRI	**brats2020-training-data**
02.	The Cancer Imaging Archive	Brain tumor MRI, Lungs CT scan, Breast cancer image, Live tumor	**https://www.cancerimagingarchive.net/**
03.	Kaggle Medical Imaging Datasets	Brain MRI images, Ultrasound images, Mammography images	**https://www.kaggle.com/datasets/andrewmvd/brain-tumor-segmentation-in-mri-brats-2015**

4.4 EXPERIMENTAL RESULTS

Segmentation means applying algorithm to divide an image into meaningful region or object. Different techniques are implemented depending on the image type and application.

Pseudo Code for Medical Image Segmentation Tool

START

1. Read grayscale image g

2. Select segmentation method from methodDrop

3. SWITCH(method)

CASE 'Otsu'

Compute threshold using Otsu method

T = graythresh(g)

Convert image into binary image

```
        seg = g > T
CASE 'Gamma'
   Set parameters
        a = 2
        b = 2
Compute Gamma probability density function
        pdf = (g^(a-1) * exp(-g/b)) / (gamma(a) * b^a)
Compute mean value of pdf
        M = mean(pdf)
Segment image
        seg = pdf > M
CASE 'Weighted Variance'
 Compute initial threshold using Otsu
        T = graythresh(g)
Create temporary segmented image
        seg = g > T
Extract object pixels
        obj = pixels where seg = 1
  Extract background pixels
        bg = pixels where seg = 0
Compute variances
        w1 = variance(obj)
        w2 = variance(bg)
 Compute weighted threshold
        thresh = (w1 * mean(obj) + w2 * mean(bg))
                / (w1 + w2)
Final segmentation
```

seg = g > thresh

CASE 'Log Normal'

Copy image

g2 = g

Replace zero pixels with epsilon value

if g2 == 0

g2 = eps

Take logarithm of image

logImg = log(g2)

Compute mean and standard deviation

mu = mean(logImg)

sigma = std(logImg)

Compute threshold

thresh = exp(mu - sigma)

Segment image

seg = g > thresh

K-MEANS CLUSTERING SEGMENTATION ALGORITHM

input :

Medical Image

Output :

Segmented Image

Step 1 :

Read and convert image into grayscale.

Step 2 :

Remove noise using median filtering and

enhance contrast.

Step 3 :

Convert image pixels into feature vector.

Step 4 :

Select number of clusters (k = 3).

Step 5 :

Apply K-Means clustering to group pixels

based on intensity similarity.

Step 6 :

Reshape clustered data into image form.

Step 7 :

Select brightest cluster as segmented region.

Step 8 :

Apply morphological operations to remove noise

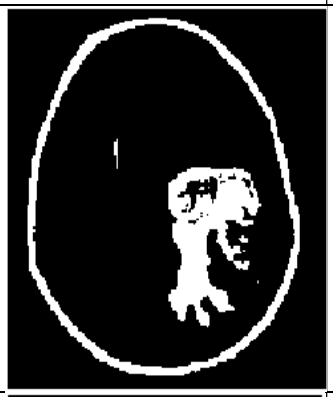
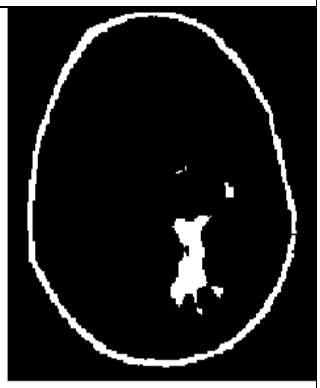
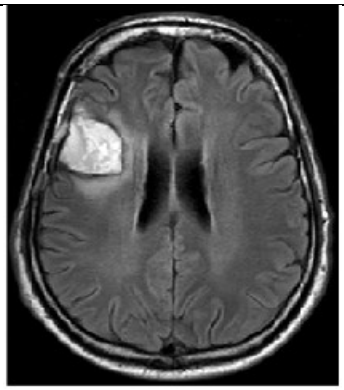
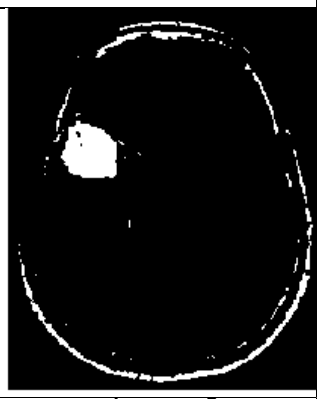
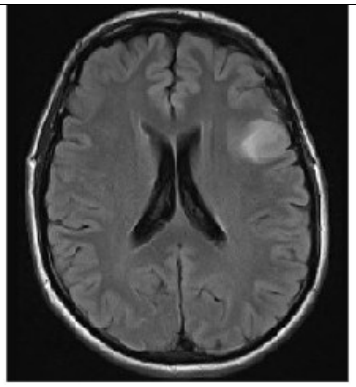
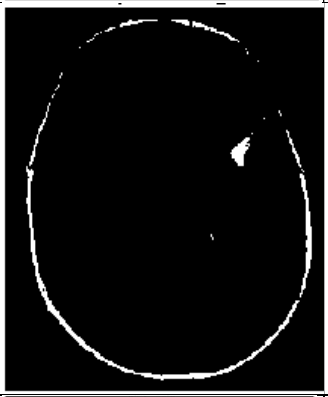
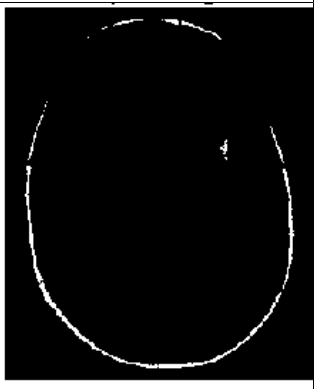
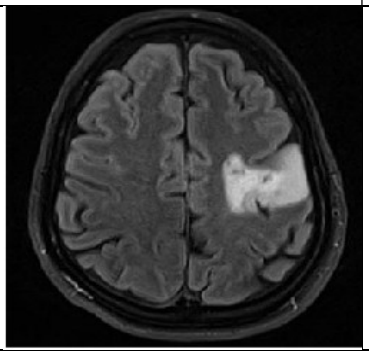
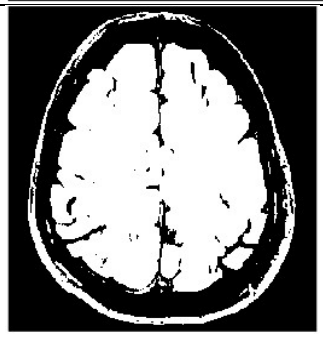
and improve segmented region.

Step 9 :

Display final segmented image.

End

Table 11. Result of DW MRI images

SR.NO.	Original image	Otsu technique	Gamma
01.			
02.			
03.			
04.			

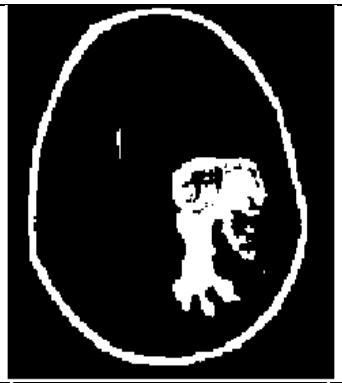
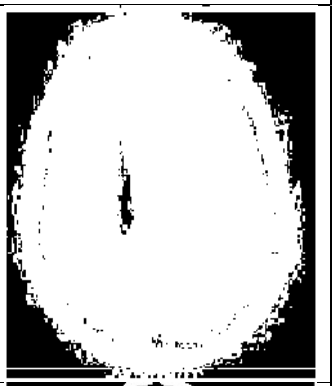
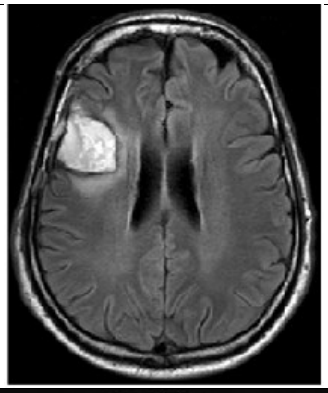
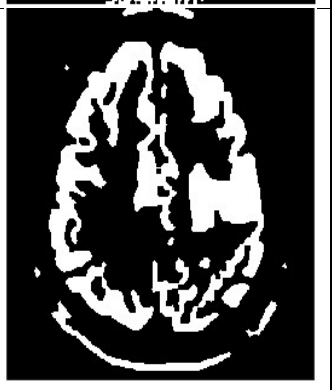
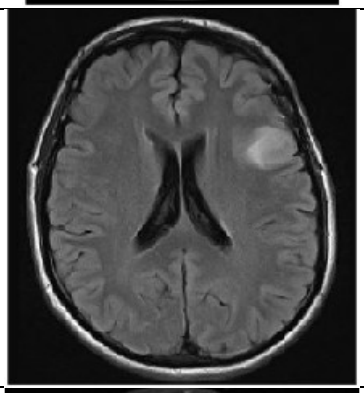
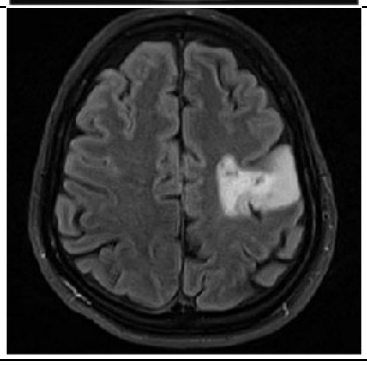
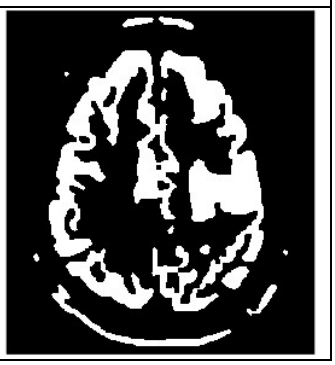
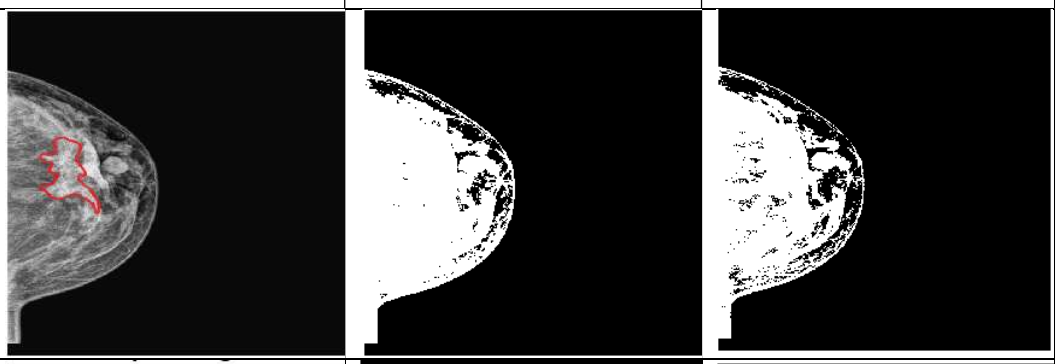
S.NO.	Original image	WOV	K-means
01.			
02.			
03			
04.			

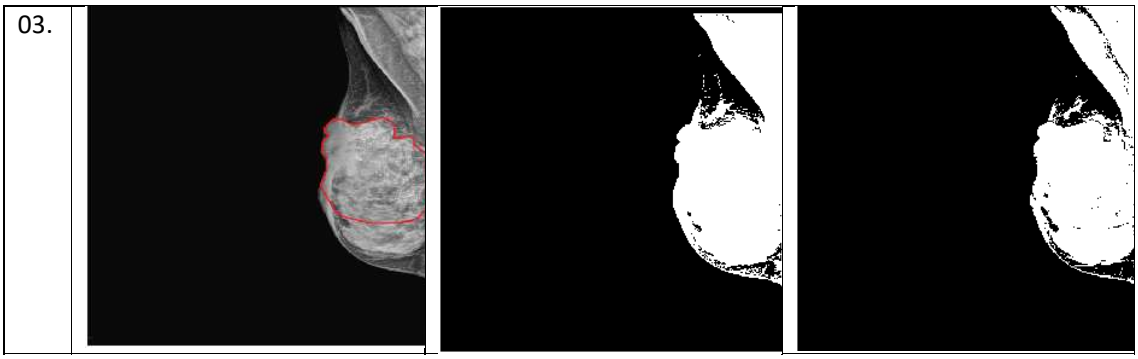
Figure 16. Result of DW MRI images

Table 12. Performance metrices extracted from result

Input image	Segment technique	Accuracy	Dice coefficient	Jaccard index	PSNR (dB)	Affected area (%)
01.	Otsu	0.6929	0.3984	0.2487	5.128	21.54
01.	Gamma	0.6875	0.3076	0.1817	5.051	15.64
01.	WOV	0.6929	0.3984	0.2487	5.128	21.54
01.	K-means	0.6796	0.4716	0.3085	4.943	25.07
02.	Otsu	0.6298	0.3294	0.1972	4.281	15.24
02.	Gamma	0.6131	0.2804	0.163	4.123	13.36
02.	WOV	0.6298	0.3294	0.1972	4.281	15.24
02.	K-means	0.6467	0.4498	0.2902	4.519	21.03
03.	Otsu	0.6502	0.265	0.1527	4.562	11.26
03.	Gamma	0.6456	0.2444	0.1392	4.505	10.57
03.	WOV	0.6502	0.265	0.1527	4.562	11.26
03.	K-means	0.56	0.3503	0.2124	3.566	32.22
04.	Otsu	0.6169	0.5871	0.4155	4.167	46.99
04.	Gamma	0.5956	0.5246	0.3555	3.932	39.26
04.	WOV	0.6169	0.5871	0.4155	4.167	46.99
04.	K-means	0.6128	0.4272	0.2716	4.121	21.80

Table 13. Result of Mammogram

S. NO.	Original image	Otsu	Gamma
01.			
02.			



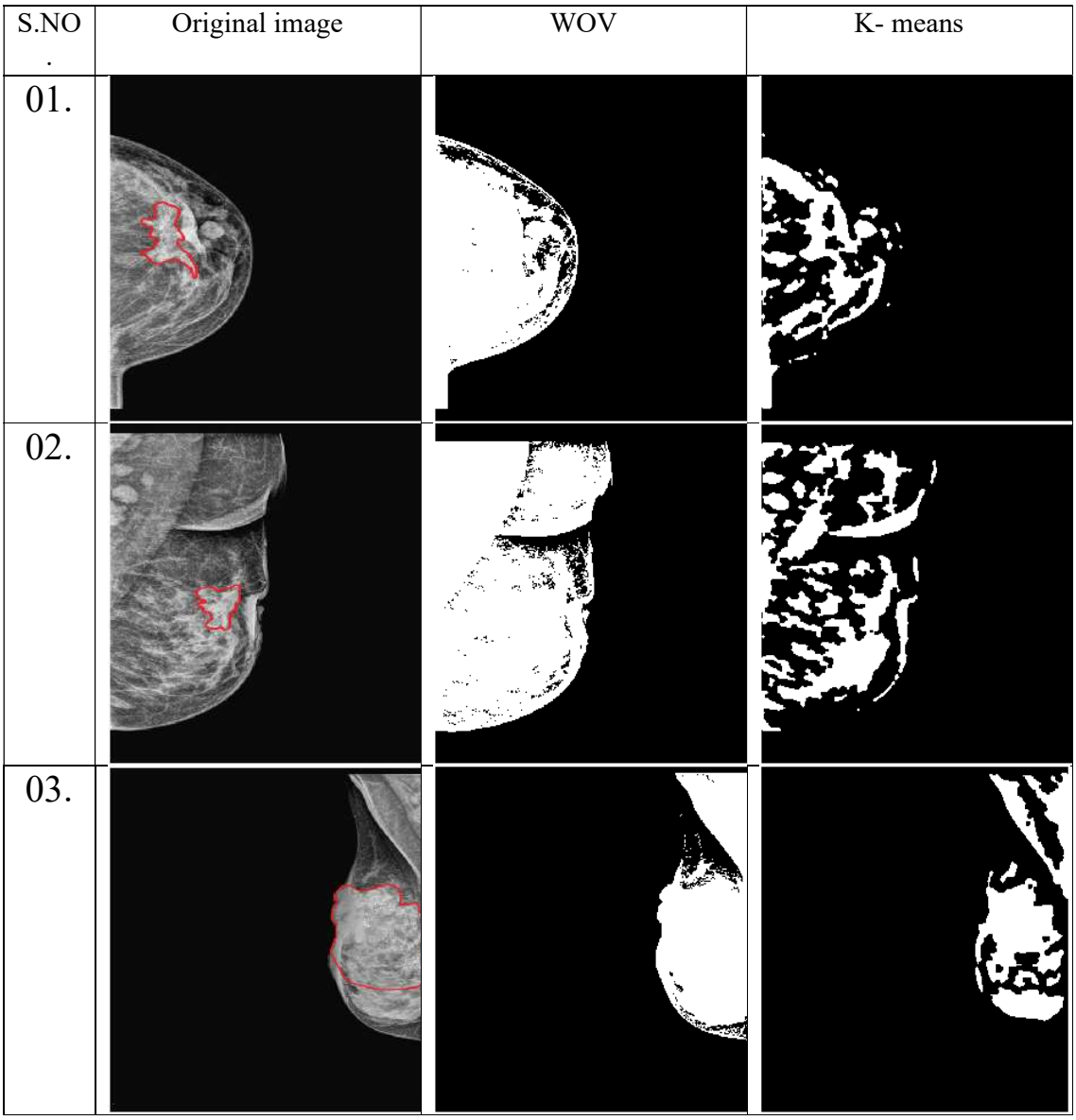


Figure 17. Result of mammogram

Table 14. Performance metrices extracted from result

Input image	Segment technique	Accuracy	Dice coefficient	Jaccard index	PSNR (dB)	Affected area (%)
01.	Otsu	0.9529	0.8864	0.796	13.27	20.18
01.	Gamma	0.9336	0.8303	0.7098	11.78	17.84
01.	WOV	0.9529	0.8864	0.796	13.27	20.18
01.	K-means	0.8518	0.4978	0.3314	8.291	8.24
02.	Otsu	0.9245	0.8943	0.8088	11.22	36.03
02.	Gamma	0.8891	0.8332	0.7141	9.552	31.03
02.	WOV	0.9245	0.8943	0.8088	11.22	36.03
02.	K-means	0.7612	0.5269	0.3576	6.219	15.09
03.	Otsu	0.9473	0.8579	0.7511	12.79	17.74
03.	Gamma	0.9347	0.8155	0.6885	11.65	16.07
03.	WOV	0.9473	0.8579	0.7511	12.79	17.74
03.	K-means	0.8832	0.5984	0.4269	9.326	9.77

Table 15. Result of Biometric image

S. NO.	Original image	Otsu	Gamma
01.			
02.			

03.			
04.			

S. NO.	Original image	WOV	K- Means
01.			
02.			

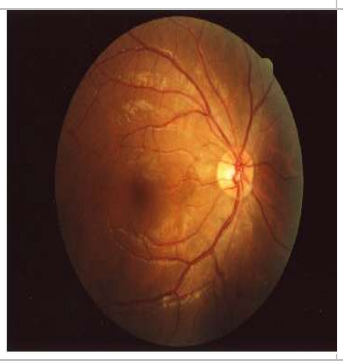
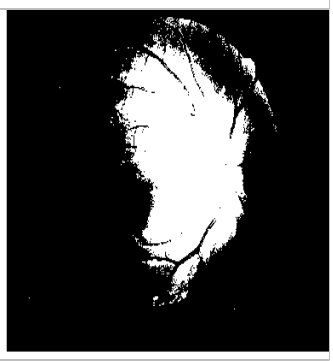
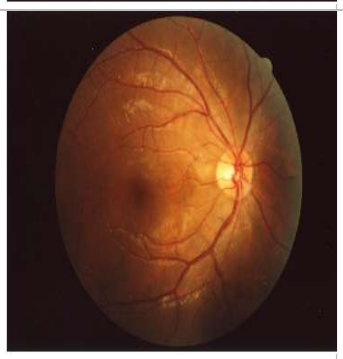
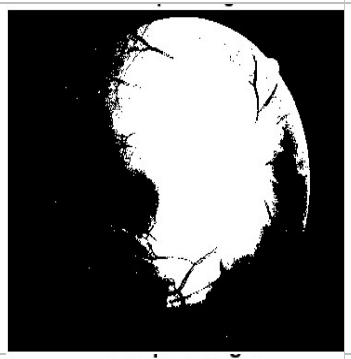
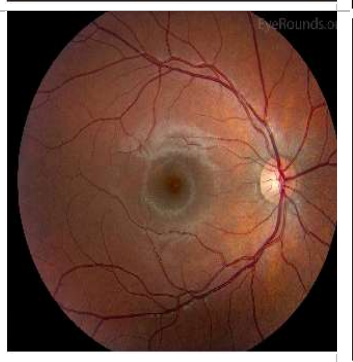
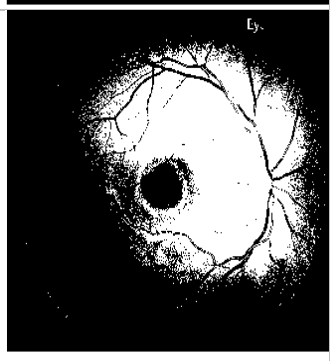
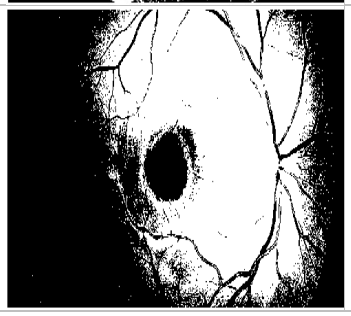


Figure 18. Result of Biometric result

Table 16. Performance metrices extracted from result

Input image	Segment technique	Accuracy	Dice coefficient	Jaccard index	PSNR (dB)	Affected area (%)
01.	Otsu	0.5543	0.7076	0.5475	3.509	67.45
01.	Gamma	0.5264	0.682	0.5174	3.246	63.94
01.	WOV	0.5543	0.7076	0.5475	3.509	67.45
01.	K-means	0.465	0.6115	0.4404	2.716	52.70
02.	Otsu	0.5527	0.708	0.548	3.494	67.83
02.	Gamma	0.5268	0.6849	0.5208	3.25	64.82
02.	WOV	0.5527	0.708	0.548	3.494	67.83
02.	K-means	0.4302	0.588	0.4164	2.443	52.94
03.	Otsu	0.5747	0.726	0.5699	3.713	76.05
03.	Gamma	0.5608	0.7136	0.5547	3.574	74.15
03.	WOV	0.5747	0.726	0.5699	3.713	76.05
03.	K-means	0.6254	0.7627	0.6165	4.264	78.73
04.	Otsu	0.5439	0.6999	0.5384	3.41	70.02
04.	Gamma	0.5255	0.6827	0.5183	3.237	67.60
04.	WOV	0.5439	0.6999	0.5384	3.41	70.02
04.	K-means	0.516	0.6622	0.495	3.151	61.31

Table 17. Result of fundus image

S.NO.	Original image	Otsu	Gamma
01.			
02.			
03.			
04			

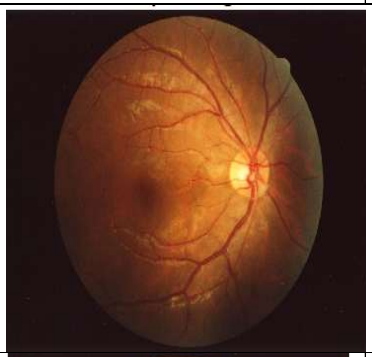
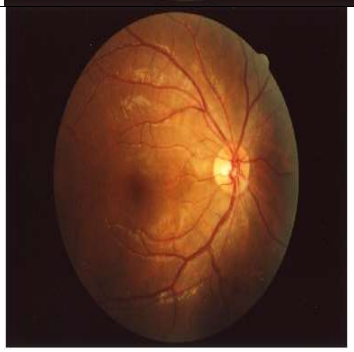
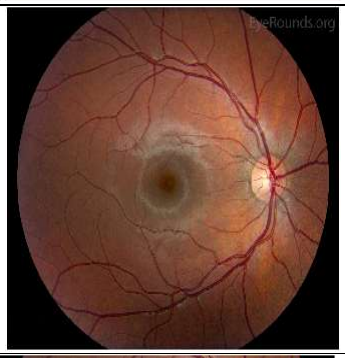
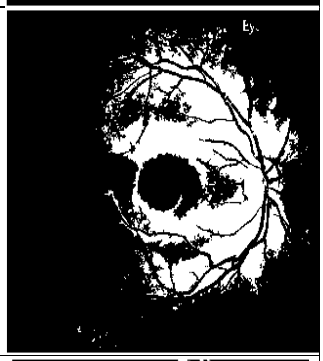
S.NO.	Original image	WOV	K- Means
01.			
02.			
03.			
04.			

Figure 19. Result of Fundus image

Table 18. Performance metrices extracted from result

Input image	Segment technique	Accuracy	Dice coefficient	Jaccard index	PSNR (dB)	Affected area (%)
01.	Otsu	0.7744	0.6834	0.519	6.467	29.24
01.	Gamma	0.7052	0.5152	0.347	5.305	18.81
01.	WOV	0.7744	0.6834	0.519	6.467	29.24
01.	K-means	0.6812	0.4439	0.2853	4.964	15.34
02.	Otsu	0.7242	0.5961	0.4246	5.595	29.26
02.	Gamma	0.6792	0.4448	0.286	4.938	18.76
02.	WOV	0.7242	05961	0.4246	5.595	29.26
02.	K-means	0.667	0.3885	0.2411	4.775	15.45
03.	Otsu	0.7387	0.7996	0.666	5.829	73.54
03.	Gamma	0.6162	0.548	0.3774	4.159	28.11
03.	WOV	0.7387	0.7996	0.666	5.829	73.54
03.	K-means	0.567	0.4354	0.2783	3.635	19.90
04.	Otsu	0.6667	0.6731	0.4674	4.917	45.00
04.	Gamma	0.6835	0.5851	0.4135	4.997	32.48
04.	WOV	0.7052	0.5152	0.347	5.305	18.81
04.	K-means	0.6757	0.5379	0.3679	4.89	26.38

CHAPTER 5

5.CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The main objective of this project was to study and compare different medical image segmentation techniques used in diagnostic applications. From the experimental analysis and literature survey, it is observed that medical image segmentation plays a significant role in modern healthcare systems.

Traditional segmentation techniques such as thresholding and edge detection are computationally simple and easy to implement. These methods provide fast results and require less computational power. However, they are sensitive to image noise, intensity variations, and low-contrast medical images. Therefore, their performance becomes limited for complex healthcare applications [1].

Region-based and clustering-based segmentation methods provide better segmentation quality by grouping similar pixels into meaningful regions. Techniques such as K-Means clustering and Fuzzy C-Means improve tissue classification and organ extraction in MRI and CT images. However, these methods require higher computational time and proper parameter selection for accurate results [13].

Watershed segmentation provides accurate boundary detection and is useful for separating connected objects in medical images. However, the major limitation of watershed segmentation is over-segmentation caused by noise and irregular gradients. Preprocessing techniques such as filtering and morphological operations are often required to improve performance [16].

The study also concludes that deep learning-based segmentation methods provide the best overall performance. Convolutional Neural Networks (CNNs), U-Net architecture, and Fully Convolutional Networks (FCNs) automatically learn image features and provide highly accurate segmentation results. These methods are particularly effective for tumor detection, organ segmentation, and disease analysis in MRI and CT images [8].

The important findings obtained from this project are:

1. Medical image segmentation improves diagnostic accuracy and assists radiologists in disease detection.
2. Thresholding techniques are simple but sensitive to noise and intensity variation.
3. Edge-based segmentation effectively detects object boundaries but fails in noisy images.
4. Region-based methods provide better continuity and object representation.

5. Clustering techniques improve tissue classification in MRI images.
6. Watershed segmentation provides accurate object separation but suffers from over-segmentation.
7. Deep learning methods provide the highest segmentation accuracy among all techniques studied.

Importance of Medical Image Segmentation in Healthcare

Medical image segmentation plays an important role in intelligent healthcare systems. It supports:

i. Early disease diagnosis
ii. Automated healthcare analysis
iii. Tumor detection and classification
iv. Surgical planning
v. Radiation therapy planning
vi. Disease monitoring and progression analysis

Segmentation techniques reduce manual workload for radiologists and improve the reliability of diagnostic systems [17].

5.2 FUTURE SCOPE

Medical image segmentation is a rapidly developing research area in healthcare technology and digital image processing. With advancements in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning, cloud computing, and medical imaging systems, future medical image segmentation techniques are expected to become more accurate, faster, and fully automated.

The future scope of medical image segmentation mainly focuses on improving diagnostic accuracy, reducing processing time, supporting intelligent healthcare systems, and enabling real-time medical analysis. Advanced segmentation systems will play an important role in disease diagnosis, robotic surgery, telemedicine, and personalized healthcare [1].

Artificial Intelligence-Based Segmentation

Artificial Intelligence (AI) and Deep Learning are transforming medical image analysis systems. Future healthcare systems will increasingly depend on intelligent segmentation algorithms capable of automatically identifying organs, tissues, tumors, and abnormal regions from medical images.

Deep learning models such as:

i. Convolutional Neural Networks (CNNs)

ii. U-Net

iii. Fully Convolutional Networks (FCNs)

iv. Mask R-CNN

v. Vision Transformers (ViTs)

are expected to provide highly accurate and automated segmentation results [8].

Real-Time Medical Image Segmentation

One of the major future developments in medical image segmentation is real-time processing. Current segmentation systems often require significant computational time, especially for high-resolution MRI and CT images. Future systems aim to provide instant segmentation outputs.

Applications of Real-Time Segmentation:

i. Robotic surgery

ii. Emergency healthcare diagnosis

iii. Live ultrasound imaging

iv. Real-time tumor detection

v. ICU patient monitoring

Real-time segmentation will help doctors make faster clinical decisions and improve surgical precision.

Hybrid Segmentation Techniques

Future research may combine multiple segmentation methods to improve overall performance. Hybrid segmentation techniques integrate the advantages of different algorithms while reducing their limitations.

Examples of Hybrid Techniques:

i. Thresholding + Watershed

ii. Clustering + Neural Networks

iii. Fuzzy Logic + Deep Learning

iv. CNN + Region Growing

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